

COMPONENTS

Universal input micro PLC

Industrial automation engineers can now achieve highly-accurate measurements for their system designs with MAXREFDES67# analogue front-end universal input micro PLC reference design from

Maxim Integrated Products. Key advantages of the product are:

Effective resolution up to 22.3 bits with temperature error as low as ± 0.1 per cent across a range of -40°C to 150°C

Beyond-the-rails signal chain reduces board space, component count and cost

Universal input accepts four different signals, namely, voltage, current, resistance temperature detector and thermocouple

Maxim Integrated Products Inc.
Website: www.maximintegrated.com

FRAM MCU

Texas Instruments has announced MSP430 ferroelectric random access memory (FRAM) microcontroller (MCU) with CapTivate technology. The new MCU offers comprehensive hardware and software features for a reliable performance in noise-challenged applications including electronic access controls, appliances, personal electronics, industrial control panels and more.

It enables designers of all skill levels to build robust human-machine interfaces with capacitive buttons, sliders, wheels or proximity sensors. It also enables multi-touch designs with plastic, glass and metal overlays of various thicknesses.

Texas Instruments
Website: www.ti.com

Wi-Fi module

ESP-WROOM-02 is a 32-bit low-power MCU Wi-Fi module that integrates TCP/IP network stacks and includes 10-bit analogue-to-digital converter



with HSPI, SDIO, UART, PWM, I2C and I2S interfaces. It can be easily integrated into resource-limited devices due to its small form factor of only 18mm x 20mm.

In addition to ESP8266EX chipset, the module also integrates an SPI flash with 32Mbits capacity in SOP8-150mil. The SPI flash is incredibly useful for storing user programs and firmware. ESP8266EX offers a complete and self-contained Wi-Fi networking solution; it can be used to host the application or to offload Wi-Fi networking functions from another application processor.

Uchi Embedded Solutions
Website: www.uchiembedded.co.in

Transceiver IC

ROHM has announced a transceiver integrated circuit (IC) compatible with the new low-speed, low-cost CXPI communication protocol for use in body control applications, including steering switch, AC and instrument panel systems.

BD41000FJ-C enables multiplexing between ECUs, contributing to decreased vehicle weight and improved fuel efficiency by reducing the number and size of wiring harnesses.

In addition, low quiescent current



(3pA typ.) and high ESD resistance ($\pm 8\text{kV}$, IEC61000-4-2) make it possible to achieve low-power, high-reliability CXPI communication.

ROHM Co. Ltd
Website: www.rohm.com

Insulated-gate bipolar transistor

Renesas Electronics has announced the new RJP65T54DPM-E0, a power semiconductor device for use in air-conditioner power factor correction circuits that employ partial switching, which are widely used in mid-to low-end air-conditioners.

Renesas Electronics Corp.
Website: www.renesas.com

DC/DC converter

The R3 DC/DC converter features ultra-low standby consumption, high efficiency in all input ranges, large



capacitive loads and multiple protections (input under-voltage protection, output over-voltage protection, over-current protection and short-circuit protection) that allows it to be used in most applications.

The average efficiency of R3 reaches 80 per cent at 10 per cent to 100 per cent load. Even if the load is lower than ten per cent, light-load efficiency of R3 exceeds 78 per cent and standby power consumption is low to 0.12W, which solves the temperature-rise problem perfectly when operating in light-load conditions for a long time and enhances reliability. In addition, these offer 4:1 wide input voltage range, 1500V DC/3000V